

HIERARCHICAL CLUSTERING

Introduction

An unsupervised machine learning algorithm that builds a multilevel hierarchy of clusters by either agglomerative approach or divisive approach, based on a chosen distance metric and linkage criterion.

The result is a dendrogram, a tree-like structure that represents nested groupings of data and their proximity relationships.

Types

- **Agglomerative:** Merges closest clusters bottom-up. Most common.
- **Divisive:** Splits data top-down. More complex.
- **BIRCH:** Builds a tree of clusters using CF (Clustering Feature) trees. Fast, good for large datasets.

Similarity Metrics

These determine how the distance between data points is measured

- **Euclidean:** Straight-line distance. Great for numeric data.
- **Manhattan:** Grid-based distance. Useful for discrete steps.
- **Cosine:** Angle between vectors. Ideal for text or high-dim data.

Linkage Methods

Determines how distances between clusters are calculated during merging.

Single Linkage



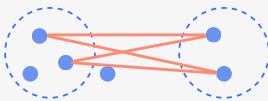
- Closest points between clusters.
- Can form long chains.

Complete Linkage



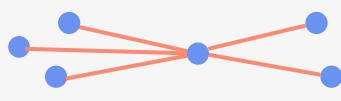
- Farthest points between clusters.
- Tighter, rounder clusters.

Average Linkage

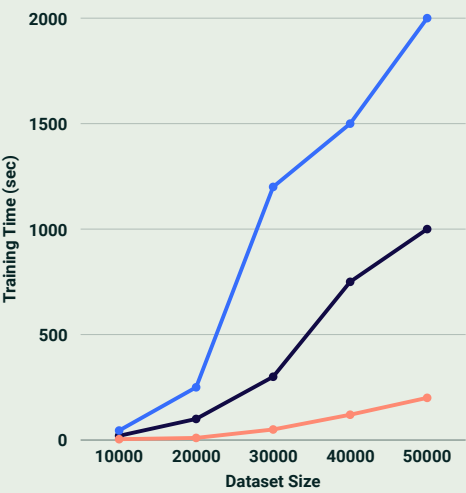


- Mean distance across clusters.
- Balanced structure.

Ward's Method



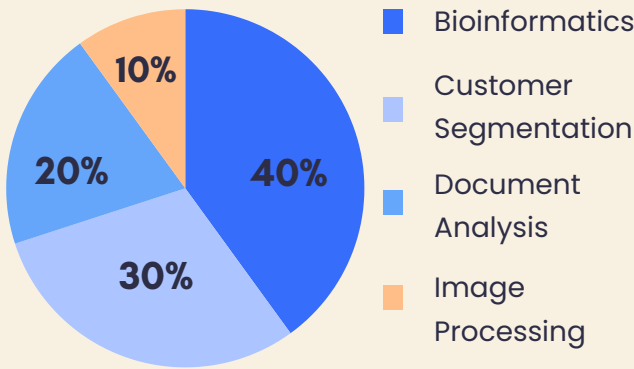
- Minimizes within-cluster variance.
- Compact, spherical shapes.



Algorithm Performance

- **Agglomerative:** $O(n^3)$ worst-case.
- **Divisive:** $O(2^n)$ in naive form.
- **BIRCH:** $O(n)$ (efficient and fast)

Applications



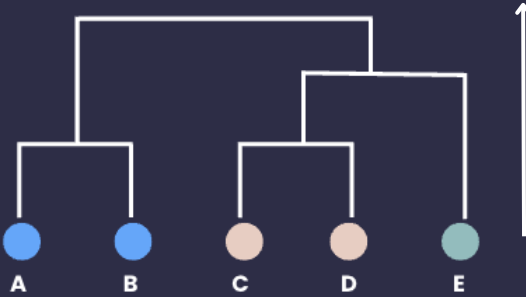
Agglomerative Clustering

Agglomerative clustering is a bottom-up hierarchical method where each data point starts as its own individual cluster. These clusters are then repeatedly merged based on similarity until a single, unified cluster remains.

Steps

1. **Start** – Each point is a cluster
2. **Measure** – Calculate distances
3. **Merge** – Join closest clusters
4. **Update** – Recalculate distances
5. **Repeat** – Until one cluster remains

Dendrogram Visualization



Advantages

- Doesn't require specifying the number of clusters in advance.
- Provides an interpretable hierarchical structure.
- Flexible to different distance/linkage methods.

Disadvantages

- Computationally intensive for large datasets.
- Sensitive to noise and outliers.
- Early merging decisions are not reversible.